Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Linear Regression

Name : Aayansh Niranjan Date : 15/11/2025
Reg_No.: 20253004

Scenario 1: Predicting House Prices from Size

You are a data analyst working for a real estate agency. The agency wants to help clients estimate the selling price of their homes based on a simple feature: the square footage of the house. You have been given a small dataset of recently sold houses, containing their size in square feet and their final selling price.

Your task is to:

- Build a simple linear regression model that predicts the house price based on its size.
- Use the model to predict the price of a house that is 2,100 square feet.
- Explain the relationship (the model's slope) between the size of a house and its price.

A1. CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
sns.set(style="whitegrid")

data = pd.read_csv('house.csv')
X = data[['Size']]
y = data['Price']

model = LinearRegression()
model.fit(X, y)

size_to_predict = pd.DataFrame({'Size': [2100]})
predicted_price = model.predict(size_to_predict)
print(f"Predicted price for a 2100 sq ft house: ${predicted_price[0]:,.2f}")

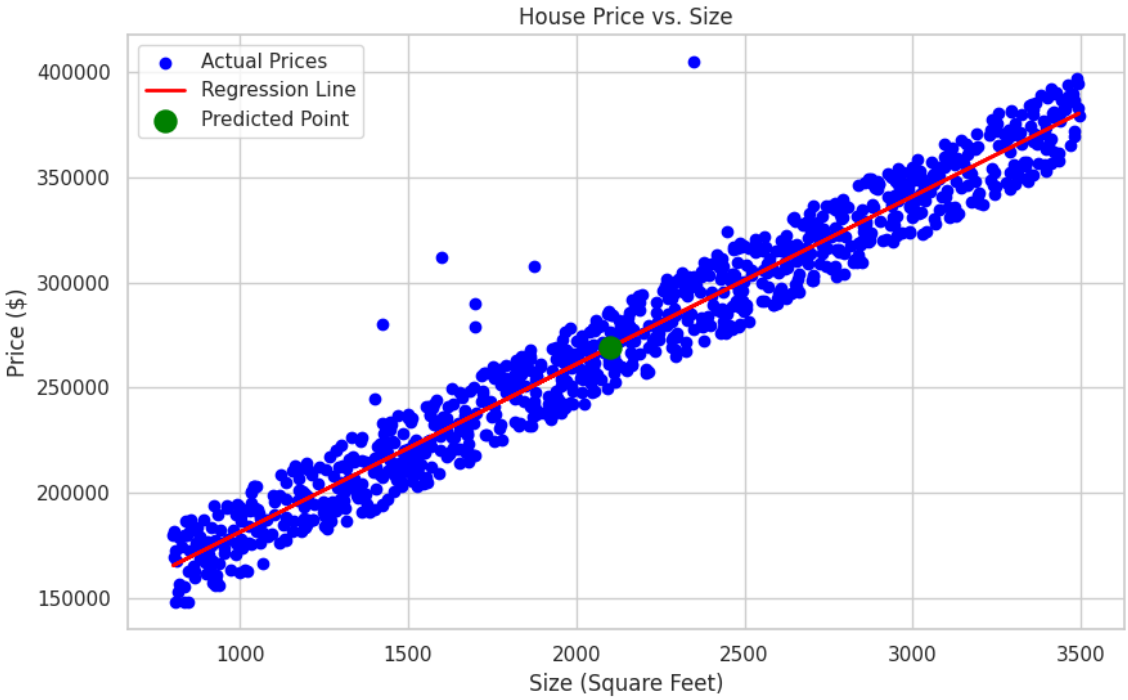
slope = model.coef_[0]
intercept = model.intercept_
print(f"Model Slope (Price per sq ft): ${slope:.2f}")
print(f"Model Intercept: ${intercept:,.2f}")
print("\nThis means for each additional square foot, the price is expected to increase")

plt.figure(figsize=(10, 6))
plt.scatter(X, y, color='blue', label='Actual Prices')
plt.plot(X, model.predict(X), color='red', linewidth=2, label='Regression Line')
plt.scatter(size_to_predict, predicted_price, color='green', s=150, zorder=5, label='Predicted Point')
plt.title('House Price vs. Size')
plt.xlabel('Size (Square Feet)')
plt.ylabel('Price ($)')
plt.legend()
plt.grid(True)
plt.savefig("house_plot.png")
print("Plot saved as house_plot.png")
print("-" * 40 + "\n")
```

OUTPUT:

```
(venv) aayansh207@aayansh-PC:~/Desktop/MNNIT/Linear_Regression$ python3 p1.py
Predicted price for a 2100 sq ft house: $269,143.90
Model Slope (Price per sq ft): $79.80
Model Intercept: $101,571.23

This means for each additional square foot, the price is expected to increase
Plot saved as house_plot.png
----------------------------------------
```



Scenario 3: Predicting Petal Length from Petal Width

A botanist has observed a strong correlation between the petal width and petal length of Iris flowers. They want to create a model to quantify this relationship and predict the length of a petal if they only know its width.

Your task is to:

- Build a simple linear regression model using the Iris dataset to predict petal length (dependent variable) based on petal width (independent variable).
- Use the trained model to predict the petal length for a flower with a petal width of 1.5 cm.
- Visualize the relationship with a scatter plot and the regression line.

A2. CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
sns.set(style="whitegrid")
data=pd.read_csv("iris.csv")
X=data[['petal_length']]
y=data['petal_width']

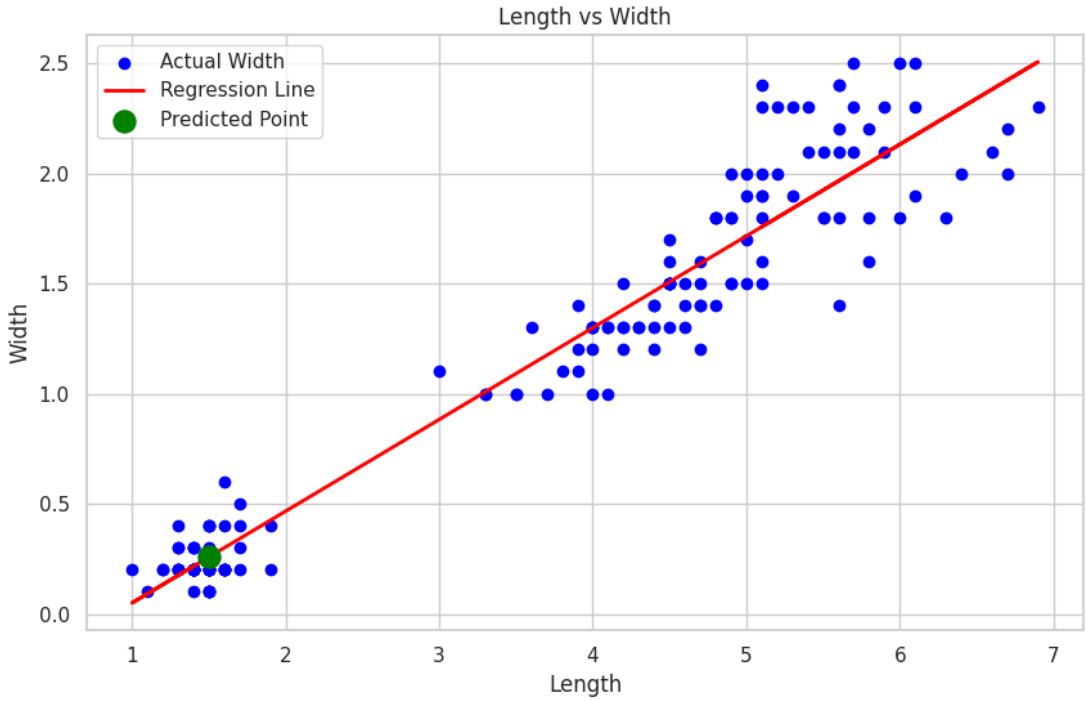
model=LinearRegression()
model.fit(X,y)

predicted=model.predict(pd.DataFrame({'petal_length': [1.5]}))
print("Predicted Width = ",predicted)

plt.figure(figsize=(10, 6))
plt.scatter(X,y,color='blue',label='Actual Width')
plt.plot(X,model.predict(X),color='red',linewidth=2,label="Regression Line")
plt.scatter(pd.DataFrame({'petal_length':
[1.5]}),predicted,s=150,zorder=5,color='green',label='Predicted Point')
plt.title('Length vs Width')
plt.xlabel('Length')
plt.ylabel('Width')
plt.legend()
plt.grid(True)
plt.savefig("Iris_plot.png")
print("Plot saved as Iris_plot.png")
```

OUTPUT:

```
(venv) aayansh207@aayansh-PC:~/Desktop/MNNIT/Linear_Regression$ python3 p3.py
Predicted Width =  [0.25811465]
Plot saved as Iris_plot.png
```



Scenario 4: Predicting Sepal Length from Sepal Width

Following the success of the petal model, the botanist now wants to see if a similar predictive relationship exists
between the sepal measurements. They are skeptical that sepal width is a good predictor for sepal length.

Your task is to:

- Build a simple linear regression model to predict sepal length (dependent variable) based on sepal width(independent variable).
- Evaluate the model's performance using the R-squared value.
- Based on the R-squared value, conclude whether sepal width is a good predictor for sepal length.

A3. CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
sns.set(style="whitegrid")
data=pd.read_csv("iris.csv")
X=data[['sepal_length']]
y=data['sepal_width']

model=LinearRegression()
model.fit(X,y)

y_pred=model.predict(X)
r2 = r2_score(y, y_pred)
print("R2 Score is : ",r2)
plt.figure(figsize=(10, 6))
plt.scatter(X,y,color='blue',label='Actual Width')
plt.plot(X,model.predict(X),color='red',linewidth=2,label="Regression Line")
plt.title('Length vs Width')
plt.xlabel('Length')
plt.ylabel('Width')
plt.legend()
plt.grid(True)
plt.savefig("Iris_plot2.png")
print("Plot saved as Iris_plot2.png")
```

OUTPUT:

```
(venv) aayansh207@aayansh-PC:~/Desktop/MNNIT/Linear_Regression$ python3 p4.py
R2 Score is :  0.0119616328347677
Plot saved as Iris_plot2.png
```

